Date: 11/07/2025

Lab Practical #06:

Study Client-Server Socket programming - TCP & UDP

Practical Assignment #06:

1. **Write a C/Java code for TCP Server-Client Socket Programming.**
2. **Write a C/Java code for UDP Server-Client Socket Programming.**

1. For TCP Server-Client:

TCP Server Program:

```java
// Demonstrating Server-side Programming
import java.net.*;
import java.io.*;
class Server {

    // Initialize socket and input stream
    private Socket s = null;
    private ServerSocket ss = null;
    private DataInputStream in = null;

    // Constructor with port
    public Server(int port) {
        // Starts server and waits for a connection
        try
        {
            ss = new ServerSocket(port);
            System.out.println("Server started");

            System.out.println("Waiting for a client ...");

            s = ss.accept();
            System.out.println("Client accepted");

            // Takes input from the client socket
            in = new DataInputStream(
                new BufferedInputStream(s.getInputStream()));

            String m = "";

            // Reads message from client until "Over" is sent
            while (!m.equals("Over"))
            {
                try{
                    m = in.readUTF();
                    System.out.println(m);}
                catch(IOException i)
                {System.out.println(i);}
            }
            System.out.println("Closing connection");
            // Close connection
            s.close();
            in.close();}
        catch(IOException i)
        {
            System.out.println(i);
        }}}
public class TCP_Server{
    public static void main(String args[])
    {Server s = new Server(5000);}}
```

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TCP Client Program:

```java
// Demonstrating Client-side Programming
import java.io.*;
import java.net.*;
class Client {

    // Initialize socket and input/output streams
    private Socket s = null;
    private DataInputStream in = null;
    private DataOutputStream out = null;

    // Constructor to put IP address and port
    public Client(String addr, int port){
        // Establish a connection
        try {
            s = new Socket(addr, port);
            System.out.println("Connected");

            // Takes input from terminal
            in = new DataInputStream(System.in);

            // Sends output to the socket
            out = new DataOutputStream(s.getOutputStream());}
        catch (UnknownHostException u) {
            System.out.println(u);
            return;}
        catch (IOException i) {
            System.out.println(i);
            return;}

        // String to read message from input
        String m = "";

        // Keep reading until "Over" is input
        while (!m.equals("Over")) {
            try {
                m = in.readLine();
                out.writeUTF(m);}
            catch (IOException i) {
                System.out.println(i);}}

        // Close the connection
        try {
            in.close();
            out.close();
            s.close();}
        catch (IOException i) {
            System.out.println(i);}}}

public class TCP_Client{
    public static void main(String[] args) {
        Client c = new Client("127.0.0.1", 5000);}}
```

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2. For UDP Server-Client:

UDP Server Program:

```java
// Java program to illustrate Server side
// Implementation using DatagramSocket
import java.io.IOException;
import java.net.DatagramPacket;
import java.net.DatagramSocket;
import java.net.InetAddress;
import java.net.SocketException;

public class UDP_Server{
    public static void main(String[] args) throws IOException{
        // Step 1 : Create a socket to listen at port 1234
        DatagramSocket ds = new DatagramSocket(1234);
        byte[] receive = new byte[65535];
        DatagramPacket DpReceive = null;
        while (true){

            // Step 2 : create a DatgramPacket to receive the data.
            DpReceive = new DatagramPacket(receive, receive.length);

            // Step 3 : revieve the data in byte buffer.
            ds.receive(DpReceive);

            System.out.println("Client:-" + data(receive));

            // Exit the server if the client sends "bye"
            if (data(receive).toString().equals("bye")){
                System.out.println("Client sent bye.....EXITING");
                break;}

            // Clear the buffer after every message.
            receive = new byte[65535];}}

    // A utility method to convert the byte array
    // data into a string representation.
    public static StringBuilder data(byte[] a){
        if (a == null)
            return null;
        StringBuilder ret = new StringBuilder();
        int i = 0;
        while (a[i] != 0){
            ret.append((char) a[i]);
            i++;}
        return ret;}}
```

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UDP Client Program:

```java
// Java program to illustrate Client side
// Implementation using DatagramSocket
import java.io.IOException;
import java.net.DatagramPacket;
import java.net.DatagramSocket;
import java.net.InetAddress;
import java.util.Scanner;

public class UDP_Client{
    public static void main(String args[]) throws IOException{
        Scanner sc = new Scanner(System.in);

        // Step 1:Create the socket object for
        // carrying the data.
        DatagramSocket ds = new DatagramSocket();

        // InetAddress ip = InetAddress.getLocalHost();
        InetAddress ip = InetAddress.getByName("10.20.43.10");

        byte buf[] = null;

        // loop while user not enters "bye"
        while (true){
            String inp = sc.nextLine();

            // convert the String input into the byte array.
            buf = inp.getBytes();

            // Step 2 : Create the datagramPacket for sending
            // the data.
            DatagramPacket DpSend =
                    new DatagramPacket(buf, buf.length, ip, 1234);

            // Step 3 : invoke the send call to actually send
            // the data.
            ds.send(DpSend);

            // break the loop if user enters "bye"
            if (inp.equals("bye"))
                break;}}}
```